



infinlTy Online Camp JEE (Adv) 2024

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SRG GROUP

ELECTROCHEMISTRY

SUBJECT: CHEMISTRY

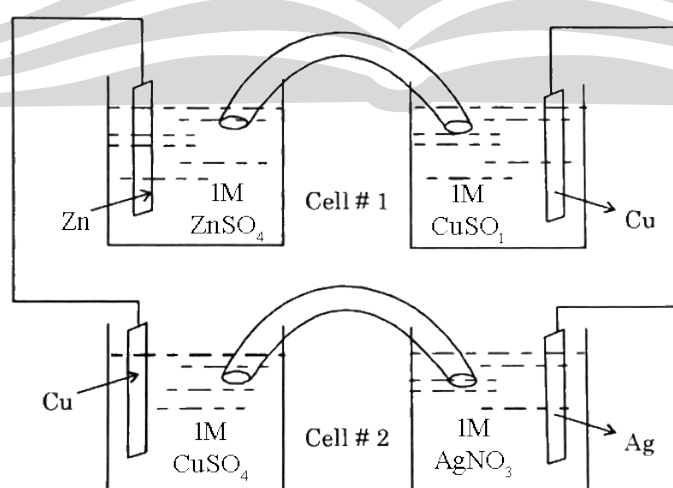
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1. The conductivity of 0.001 M Na_2SO_4 solution is $2.6 \times 10^{-4} \text{ S cm}^{-1}$ and increase to $7 \times 10^{-4} \text{ S cm}^{-1}$, when the solution is saturated with CaSO_4 . The molar conductivity of Na^+ and Ca^{2+} are 50 and $120 \text{ S cm}^2 \text{ mol}^{-1}$, respectively. Neglecting conductivity of water, the solubility product of CaSO_4 is $Y \times 10^{-x} \text{ M}^2$. Then y is.

2. In the following arrangement, find the equilibrium concentration of Cu^{+2} ion in the cathodic compartment.

[Given : $E^\circ_{\text{Zn}^{+2}/\text{Zn}} = -0.76 \text{ V}$, $E^\circ_{\text{Cu}^{+2}/\text{Cu}} = 0.34 \text{ V}$, $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$ and $\text{Antilog} \left(\frac{0.64}{0.03} \right) = 2.1 \times 10^{21}$,

$\frac{2.303RT}{F} = 0.06$, $\sqrt{\frac{6}{7}} = 0.926$ Volume of each solution is 1 L and constant.]



- (A) $9.26 \times 10^{-11} \text{ M}$ (B) $4.63 \times 10^{-11} \text{ M}$ (C) $5.28 \times 10^{-11} \text{ M}$ (D) $3.7 \times 10^{-10} \text{ M}$

Paragraph for Ques. 3 and 4

The Galvanic cell $\text{Ag}_{(s)} \left| \text{AgX}_{(s)} \right| \text{X}_{\text{aq}}^- \parallel \text{Ag}^+_{(0.05\text{M})} \left| \text{Ag}_{(s)} \right|$ has EMF 0.66 V.

[Given : $\frac{2.303RT}{F} = 0.06$]

3. Solubility product of $\text{AgX}_{(B)}$ is-
 (A) 10^{-24} (B) 10^{-18} (C) 10^{-14} (D) 10^{-10}
4. Cell EMF may increase by
 (A) Adding more $\text{Ag}_{(s)}$ at anode (B) Decreasing conc. of X at anode
 (C) Adding more $\text{Ag}_{(s)}$ at cathode (D) Increasing Ag^+ conc. at cathode

Paragraph for Ques. 5 and 6

At 298 K, the limiting molar conductivity of a weak dibasic acid is $4 \times 10^2 \text{ S cm}^2 \text{ mol}^{-1}$. At 298 K, for an aqueous solution of the acid the degree of dissociation of α and the molar conductivity is $y \times 10^2 \text{ S cm}^2 \text{ mol}^{-1}$. At 298 K, upon 10 times dilution with water, the molar conductivity of the solution becomes $3y \times 10^2 \text{ S cm}^2 \text{ mol}^{-1}$.

5. The value of α is ____.
6. The value of y is ____.
7. For a saturated solution of AgCl at 25°C , specific conductance is $3.4 \times 10^{-6} \text{ ohm}^{-1} \text{ cm}^{-1}$ and that of H_2O used for preparing the solution was $1.6 \times 10^{-6} \text{ ohm}^{-1} \text{ cm}^{-1}$. What is the solubility of AgCl ?

[Given $\Lambda_{\text{eq}}^o(\text{AgCl}) = 150 \text{ ohm}^{-1} \text{ cm}^{-1} \text{ eq}^{-1}$]

If answer of solution is $x \times 10^{-y} \text{ M}$, then find y .